Chemical Kinetics

Subject: Chemistry | NoteShare Library

CHEMICAL KINETICS

1. RATE

Rate = k[A]^m[B]^n
m+n = overall order

2. RATE LAWS

3. ARRHENIUS
k = A e^(-Ea/RT)

4. COLLISION THEORY

Effective collisions (proper orientation)

5. MECHANISMS
Elementary steps
Rate-determining step
Molecularity

6. CATALYSIS
Homogeneous
Heterogeneous
Enzymes (biological)

7. FACTORS
Concentration
Temperature (+10 C -> 2x rate)
Surface area
Catalyst
Pressure (gases)